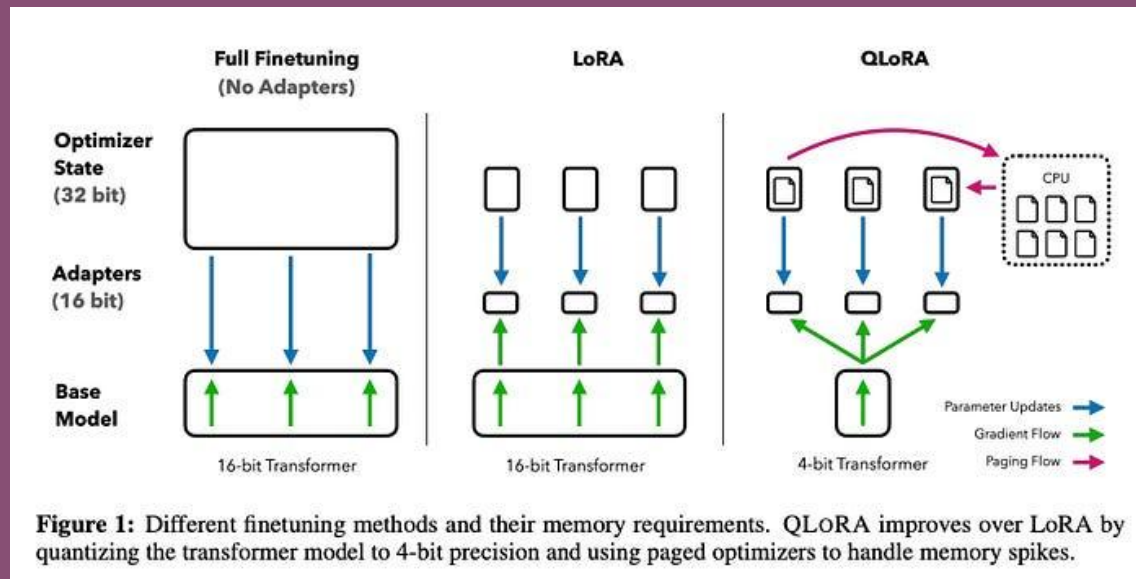
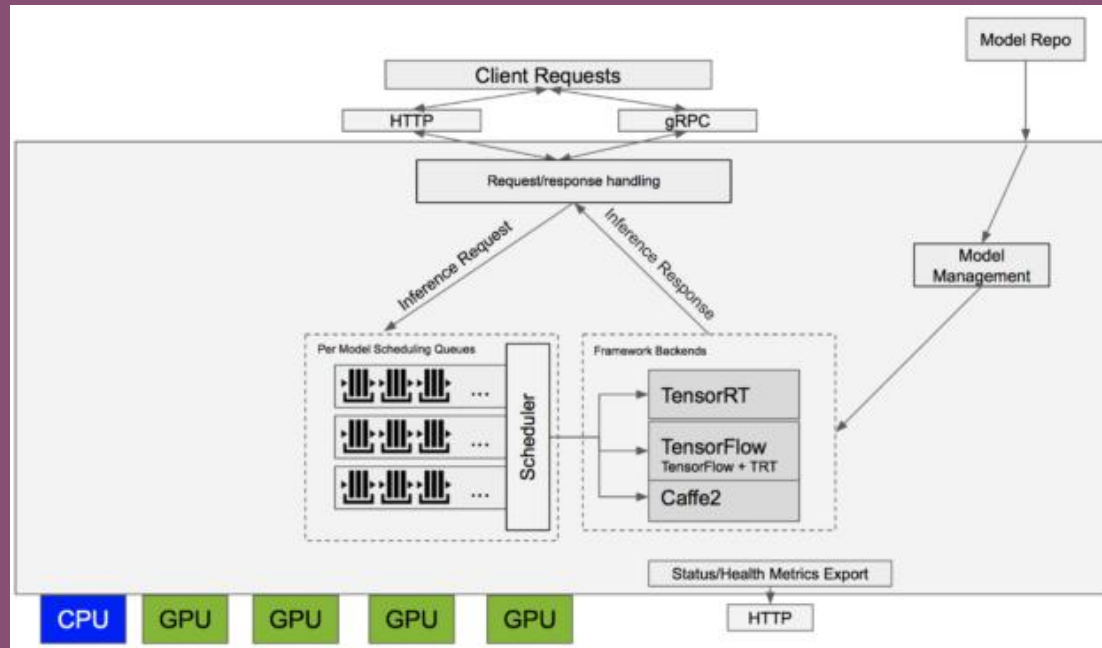


Model Shrinking: The Power of 4-bit Quantization



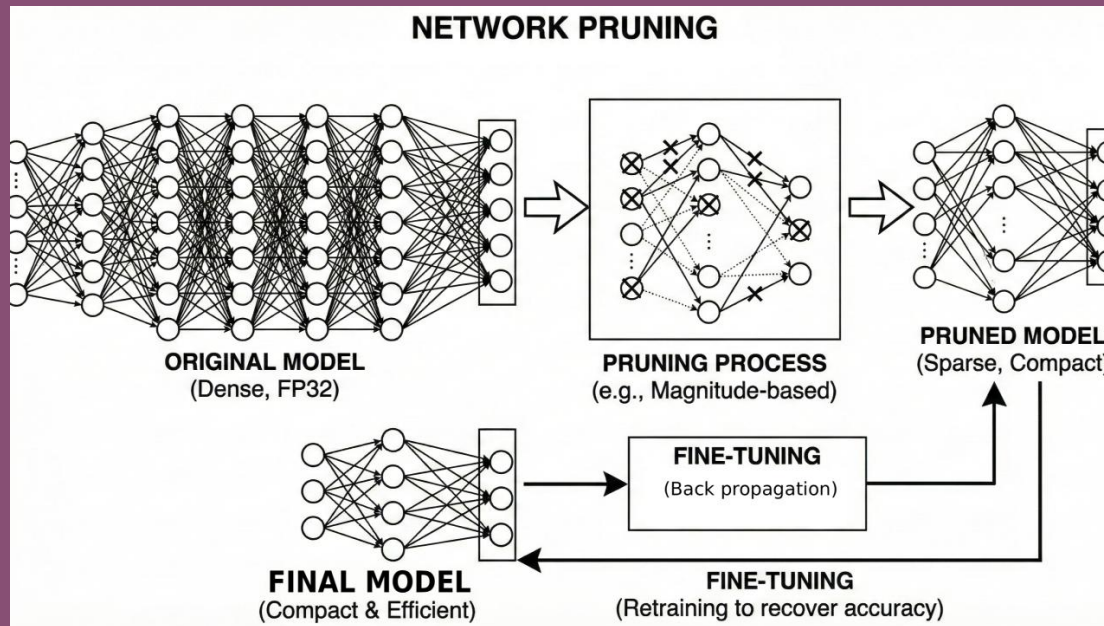
- **Quantization** reduces the precision of model weights (e.g., from 16-bit floats to 4-bit integers).
- **GGUF & EXL2:** Specialized formats that allow giant models (like a 70B parameter Llama) to run on consumer laptops by drastically reducing VRAM usage.
- This process "shrinks" the model size by 70-80% with only a minimal hit to intelligence.

Industrial Inference: ONNX & TensorRT



- **ONNX (Open Neural Network Exchange):** A universal format that allows a model trained in PyTorch to run on any hardware or framework.
- **TensorRT:** NVIDIA's optimization engine that compiles models to run at maximum speed on specific GPUs.
- These tools are essential for "Production AI" where response time (latency) must be in milliseconds.

Pruning: Trimming the Fat



- **Pruning** involves identifying and removing "useless" neurons or connections that contribute little to the final output.
- By zeroing out these weights, we make the model "sparse" and much lighter.
- Combined with quantization, pruning allows complex AI to run on "Edge" devices like smartphones and IoT sensors.